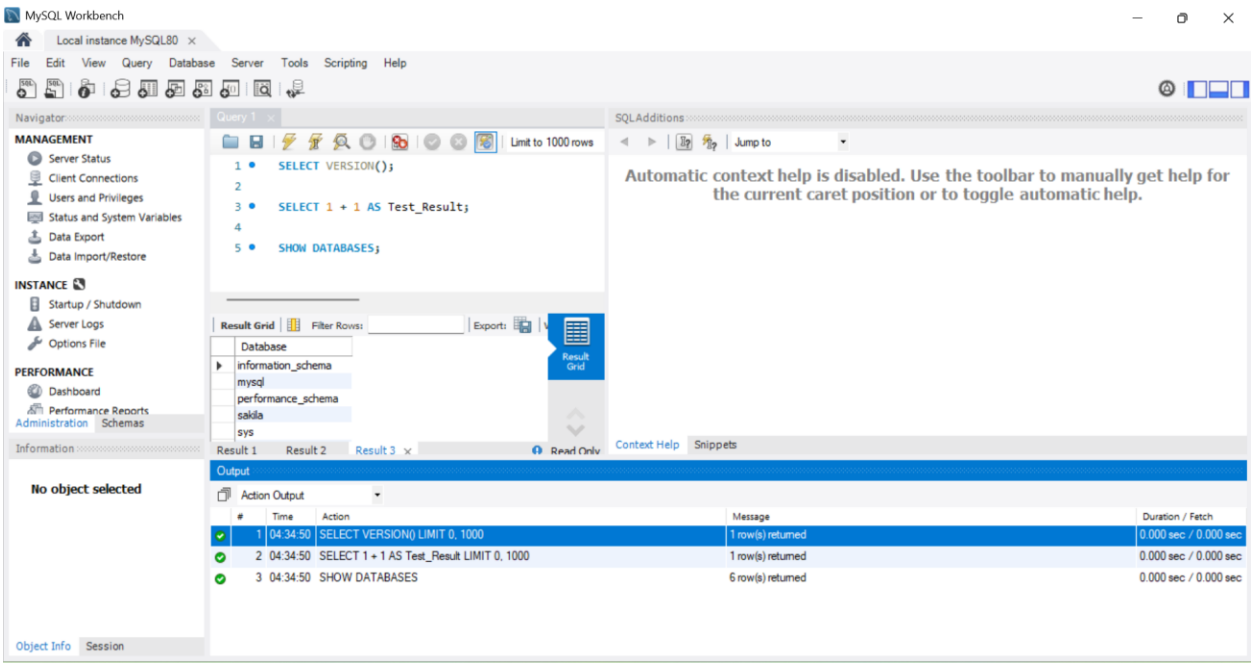
PORTFOLIO 1 – Database and Design Modelling

I began with MySQL Workbench and connected to my local MySQL server.



I used the following script to create the Table.

```
CREATE DATABASE IF NOT EXISTS UniversityManagementDB;
USE UniversityManagementDB;
-- 1. FACULTY
CREATE TABLE Faculty (
    faculty_id INT PRIMARY KEY AUTO_INCREMENT,
    faculty_name VARCHAR(100) NOT NULL UNIQUE
);
-- 2. DEPARTMENT
CREATE TABLE Department (
    department_id INT PRIMARY KEY AUTO_INCREMENT,
    department_name VARCHAR(100) NOT NULL,
    faculty_id INT NOT NULL,
    UNIQUE (department_name, faculty_id),
    FOREIGN KEY (faculty_id)
        REFERENCES Faculty(faculty_id)
);
```

```
-- 3. STUDENT
CREATE TABLE Student (
    student_id INT PRIMARY KEY AUTO_INCREMENT,
    matric_number VARCHAR(20) NOT NULL UNIQUE,
    full_name VARCHAR(120) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    department_id INT NOT NULL,
    admission_year YEAR NOT NULL,
    FOREIGN KEY (department_id)
        REFERENCES Department(department_id)
);

-- 4. LECTURER
CREATE TABLE Lecturer (
    lecturer_id INT PRIMARY KEY AUTO_INCREMENT,
    staff_number VARCHAR(20) NOT NULL UNIQUE,
    full_name VARCHAR(120) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    department_id INT NOT NULL,
    FOREIGN KEY (department_id)
        REFERENCES Department(department_id)
);

-- 5. COURSE
CREATE TABLE Course (
    course_id INT PRIMARY KEY AUTO_INCREMENT,
    course_code VARCHAR(15) NOT NULL UNIQUE,
    course_title VARCHAR(120) NOT NULL,
    credit_units INT NOT NULL,
    department_id INT NOT NULL,
    CHECK (credit_units BETWEEN 1 AND 6),
    FOREIGN KEY (department_id)
        REFERENCES Department(department_id)
);
```

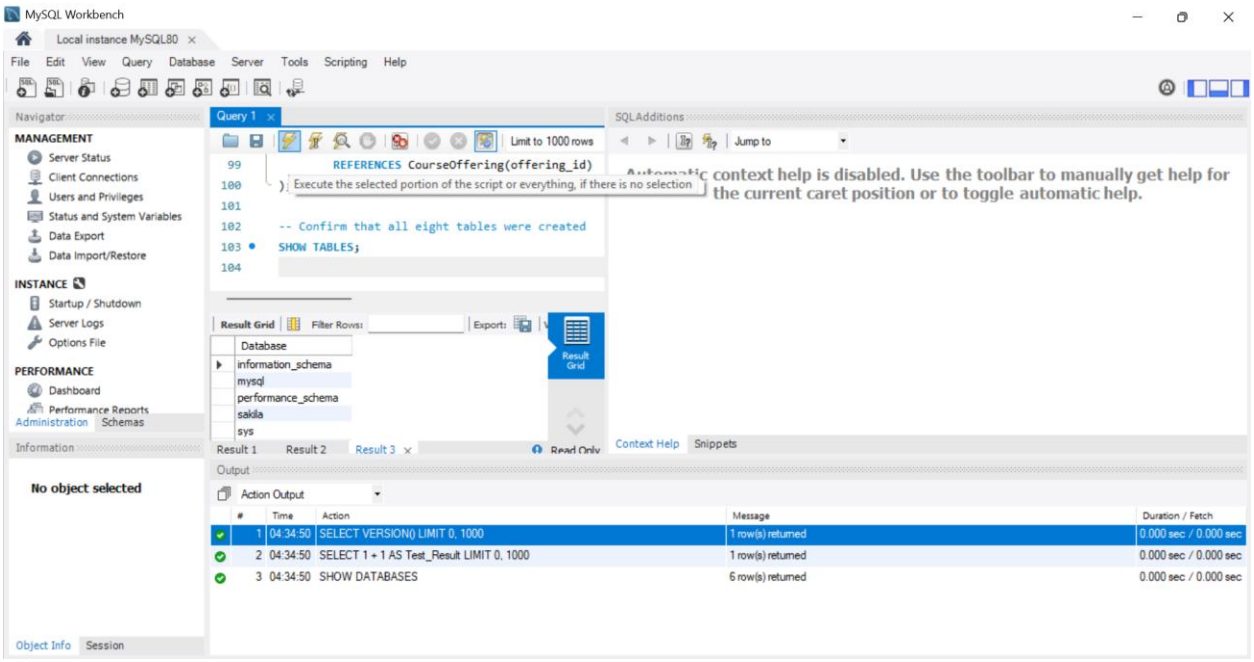
```
-- 6. SEMESTER
CREATE TABLE Semester (
    semester_id INT PRIMARY KEY AUTO_INCREMENT,
    academic_year VARCHAR(9) NOT NULL,
    semester_name VARCHAR(20) NOT NULL,
    UNIQUE (academic_year, semester_name)
);

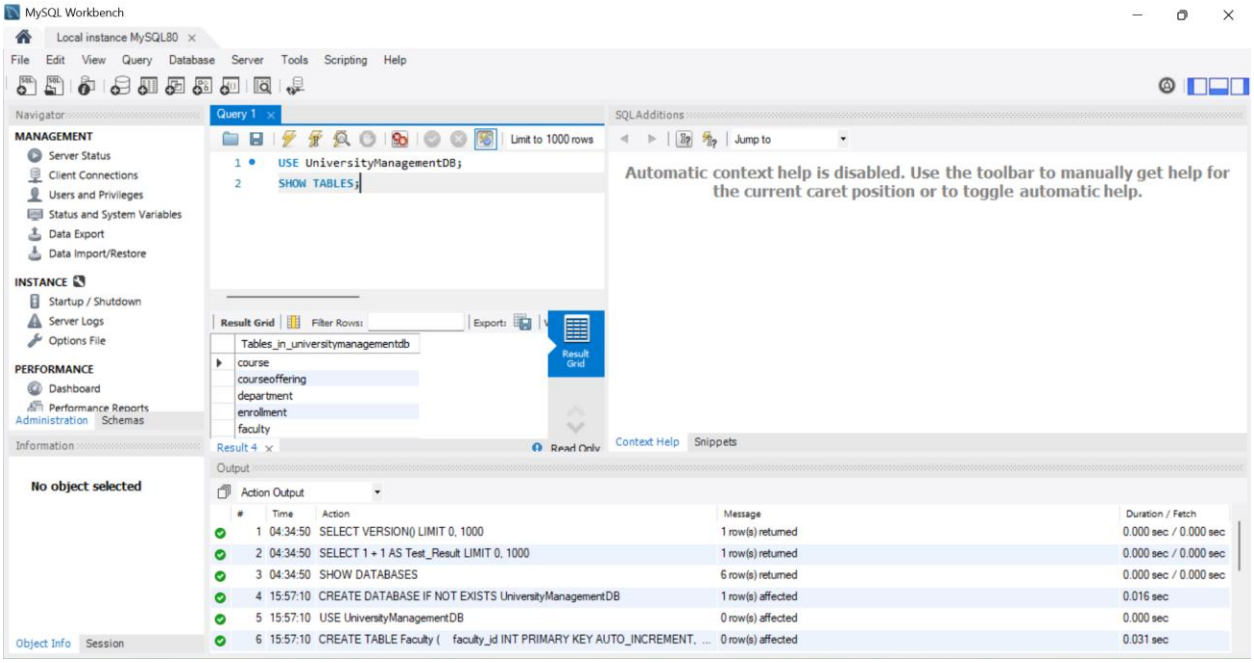
-- 7. COURSE OFFERING
CREATE TABLE CourseOffering (
    offering_id INT PRIMARY KEY AUTO_INCREMENT,
    course_id INT NOT NULL,
    lecturer_id INT NOT NULL,
    semester_id INT NOT NULL,
    section_code VARCHAR(10) NOT NULL DEFAULT 'A',
    UNIQUE (course_id, semester_id, section_code),
    FOREIGN KEY (course_id)
        REFERENCES Course(course_id),
    FOREIGN KEY (lecturer_id)
        REFERENCES Lecturer(lecturer_id),
    FOREIGN KEY (semester_id)
        REFERENCES Semester(semester_id)
);

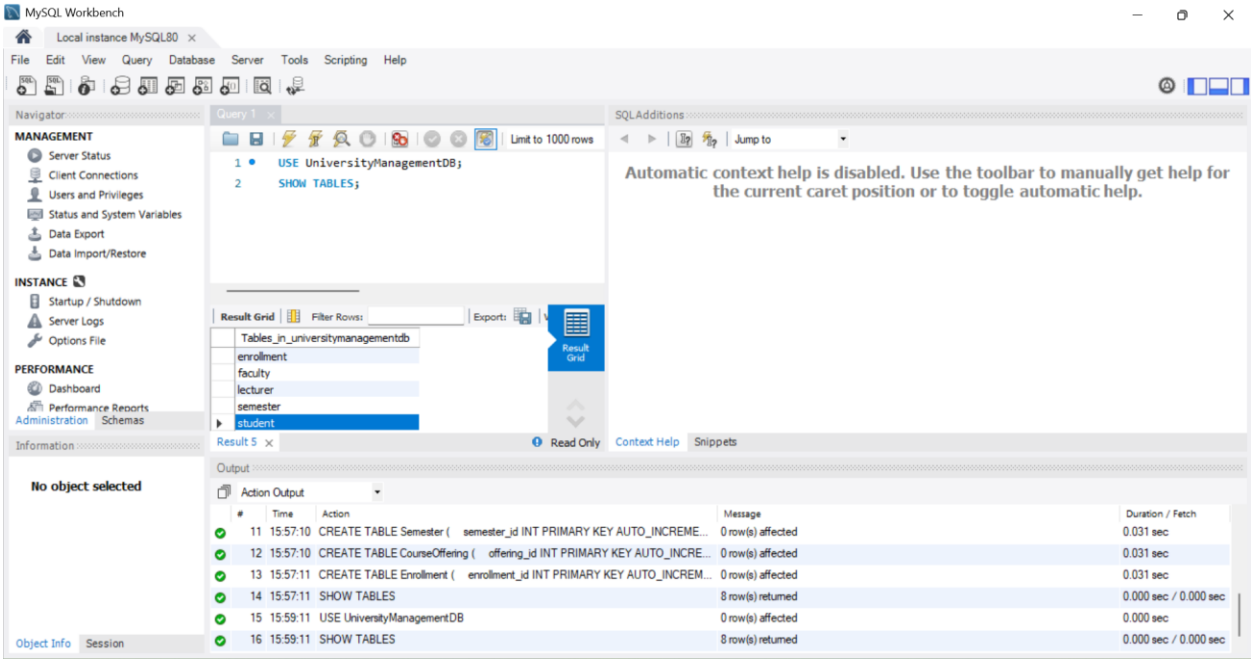
-- 8. ENROLLMENT
CREATE TABLE Enrollment (
    enrollment_id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT NOT NULL,
    offering_id INT NOT NULL,
    coursework_score DECIMAL(5,2) NOT NULL DEFAULT 0,
    exam_score DECIMAL(5,2) NOT NULL DEFAULT 0,
    total_score DECIMAL(6,2)
        GENERATED ALWAYS AS
        (coursework_score + exam_score) STORED,
```

```
enrollment_status VARCHAR(20) NOT NULL DEFAULT 'Registered',
    UNIQUE (student_id, offering_id),
    CHECK (coursework_score BETWEEN 0 AND 30),
    CHECK (exam_score BETWEEN 0 AND 70),
    CHECK (enrollment_status IN
        ('Registered', 'Completed', 'Withdrawn')),
    FOREIGN KEY (student_id)
        REFERENCES Student(student_id),
    FOREIGN KEY (offering_id)
        REFERENCES CourseOffering(offering_id)
);

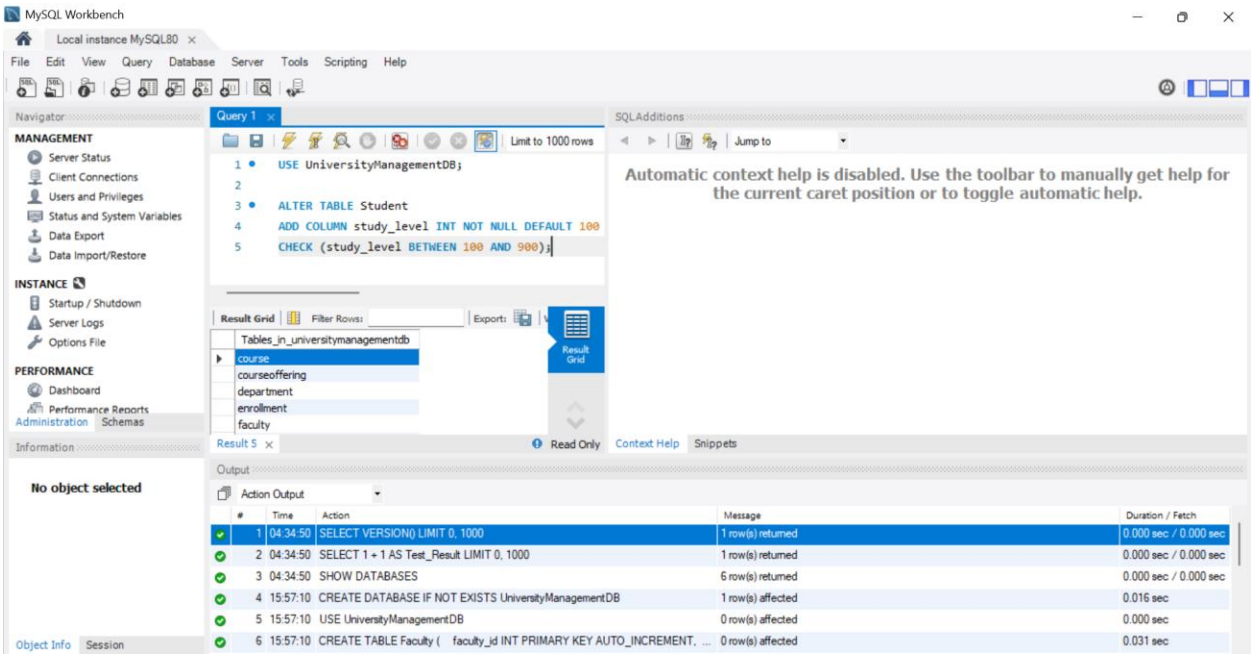
-- Confirm that all eight tables were created
SHOW TABLES;
```




MySQL Workbench
Local instance MySQL80
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Navigator
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
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Data Import/Restore
INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Reports
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 USE UniversityManagementDB;
2 SHOW TABLES;
Result Grid Filter Rows: Export:
Tables_in_universitymanagementdb
course
courseoffering
department
enrollment
faculty
Result 4
Read Only
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
1 04:34:50 SELECT VERSION() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
2 04:34:50 SELECT 1 + 1 AS Test_Result LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
3 04:34:50 SHOW DATABASES 6 row(s) returned 0.000 sec / 0.000 sec
4 15:57:10 CREATE DATABASE IF NOT EXISTS UniversityManagementDB 1 row(s) affected 0.016 sec
5 15:57:10 USE UniversityManagementDB 0 row(s) affected 0.000 sec
6 15:57:10 CREATE TABLE Faculty (faculty_id INT PRIMARY KEY AUTO_INCREMENT, ... 0 row(s) affected 0.031 sec


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
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PERFORMANCE
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Performance Reports
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 USE UniversityManagementDB;
2 SHOW TABLES;
Result Grid Filter Rows: Export:
Tables_in_universitymanagementdb
enrollment
faculty
lecturer
semester
student
Result 5
Read Only
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
11 15:57:10 CREATE TABLE Semester (semester_id INT PRIMARY KEY AUTO_INCREME... 0 row(s) affected 0.031 sec
12 15:57:10 CREATE TABLE CourseOffering (offering_id INT PRIMARY KEY AUTO_INCRE... 0 row(s) affected 0.031 sec
13 15:57:11 CREATE TABLE Enrollment (enrollment_id INT PRIMARY KEY AUTO_INCREM... 0 row(s) affected 0.031 sec
14 15:57:11 SHOW TABLES 8 row(s) returned 0.000 sec / 0.000 sec
15 15:59:11 USE UniversityManagementDB 0 row(s) affected 0.000 sec
16 15:59:11 SHOW TABLES 8 row(s) returned 0.000 sec / 0.000 sec

To Add Student Level to the Table



This Script was used to populate the table and establish Relationships between the Records.

```
USE UniversityManagementDB;

-- 1. FACULTIES

INSERT INTO Faculty (faculty_id, faculty_name) VALUES

(1, 'Faculty of Computing'),

(2, 'Faculty of Management Sciences');


-- 2. DEPARTMENTS

INSERT INTO Department

(department_id, department_name, faculty_id) VALUES

(1, 'Information Technology', 1),

(2, 'Computer Science', 1),

(3, 'Business Administration', 2);


-- 3. STUDENTS

INSERT INTO Student

(student_id, matric_number, full_name, email,

 department_id, admission_year, study_level)

VALUES
```

```
(1, 'MIT001', 'John Okafor', 'john.okafor@example.com',
 1, 2023, 800),
(2, 'MIT002', 'Mary Adeyemi', 'mary.adeyemi@example.com',
 2, 2024, 700),
(3, 'MIT003', 'David Eze', 'david.eze@example.com',
 1, 2023, 800);

-- 4. LECTURERS
INSERT INTO Lecturer
(lecturer_id, staff_number, full_name, email, department_id)
VALUES
(1, 'STAFF001', 'Dr. James Obi', 'james.obi@example.com', 1),
(2, 'STAFF002', 'Dr. Grace Bello', 'grace.bello@example.com', 2);

-- 5. COURSES
INSERT INTO Course
(course_id, course_code, course_title, credit_units, department_id)
VALUES
(1, 'MIT8103', 'Advanced Database Systems', 3, 1),
(2, 'MIT8105', 'Advanced Computer Networks', 2, 1),
(3, 'MIT8106', 'Software Engineering', 3, 2);

-- 6. SEMESTER
INSERT INTO Semester
(semester_id, academic_year, semester_name)
VALUES
(1, '2026/2027', 'First Semester');

-- 7. COURSE OFFERINGS
INSERT INTO CourseOffering
(offering_id, course_id, lecturer_id, semester_id, section_code)
VALUES
(1, 1, 1, 1, 'A'),
(2, 2, 1, 1, 'A'),
(3, 3, 2, 1, 'A');
```

-- 8. ENROLLMENTS

INSERT INTO Enrollment

(enrollment_id, student_id, offering_id,

coursework_score, exam_score, enrollment_status)

VALUES

(1, 1, 1, 25, 60, 'Completed'),

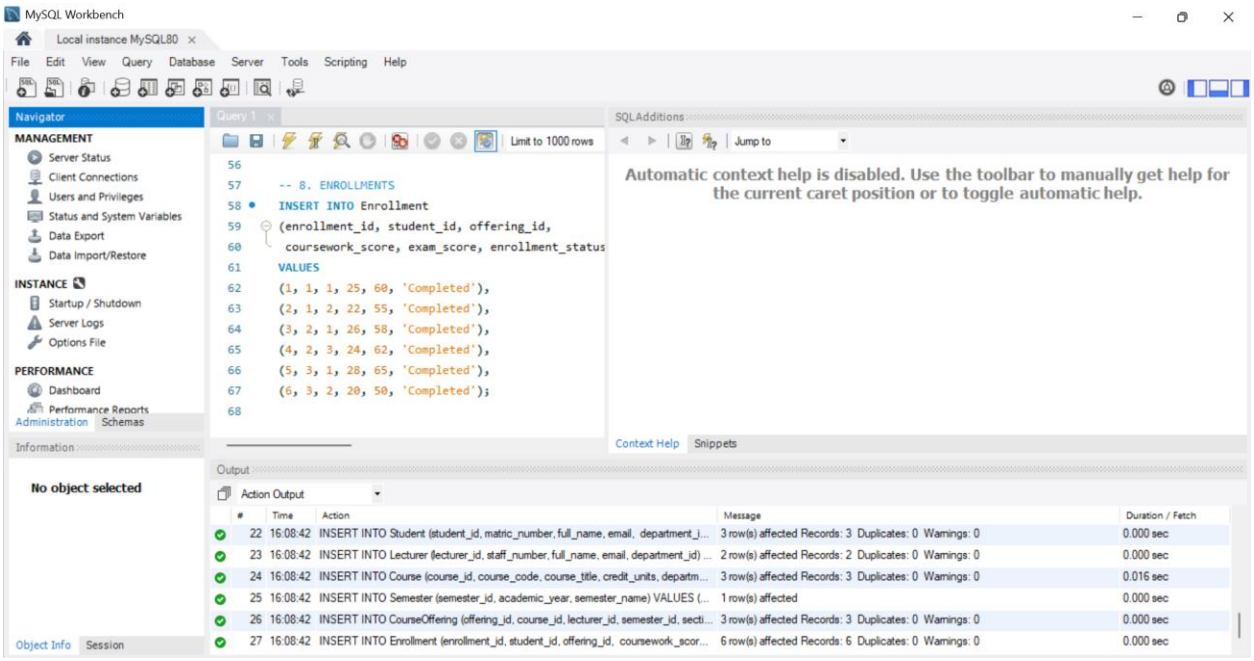
(2, 1, 2, 22, 55, 'Completed'),

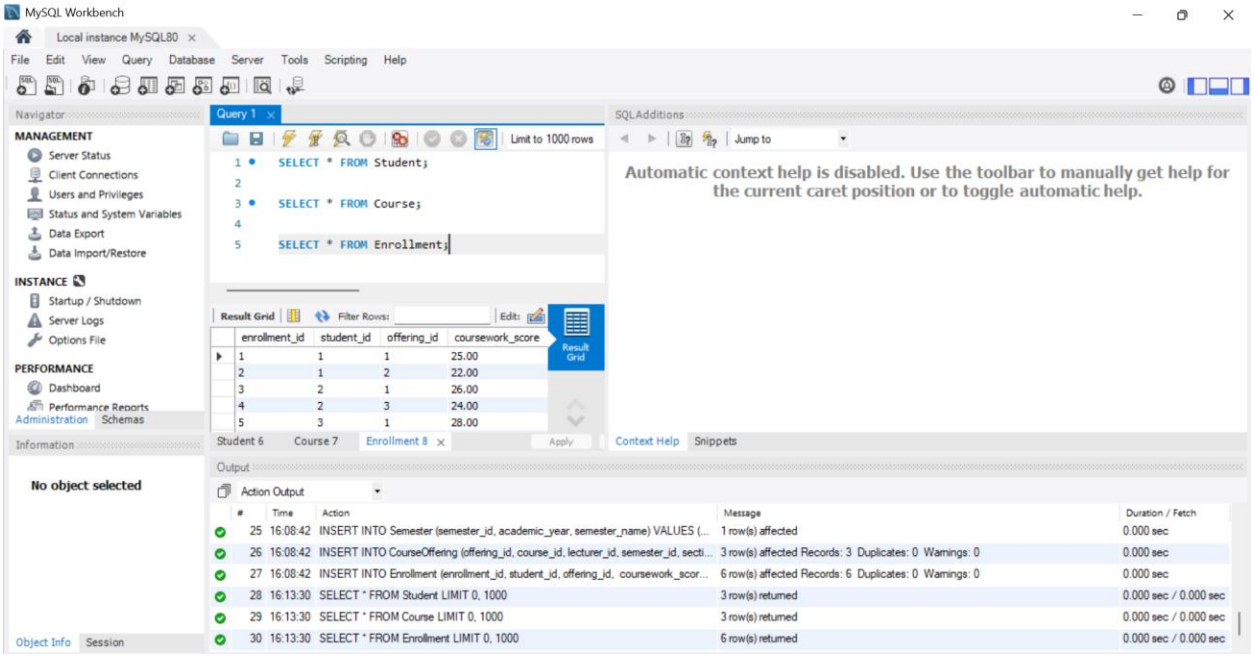
(3, 2, 1, 26, 58, 'Completed'),

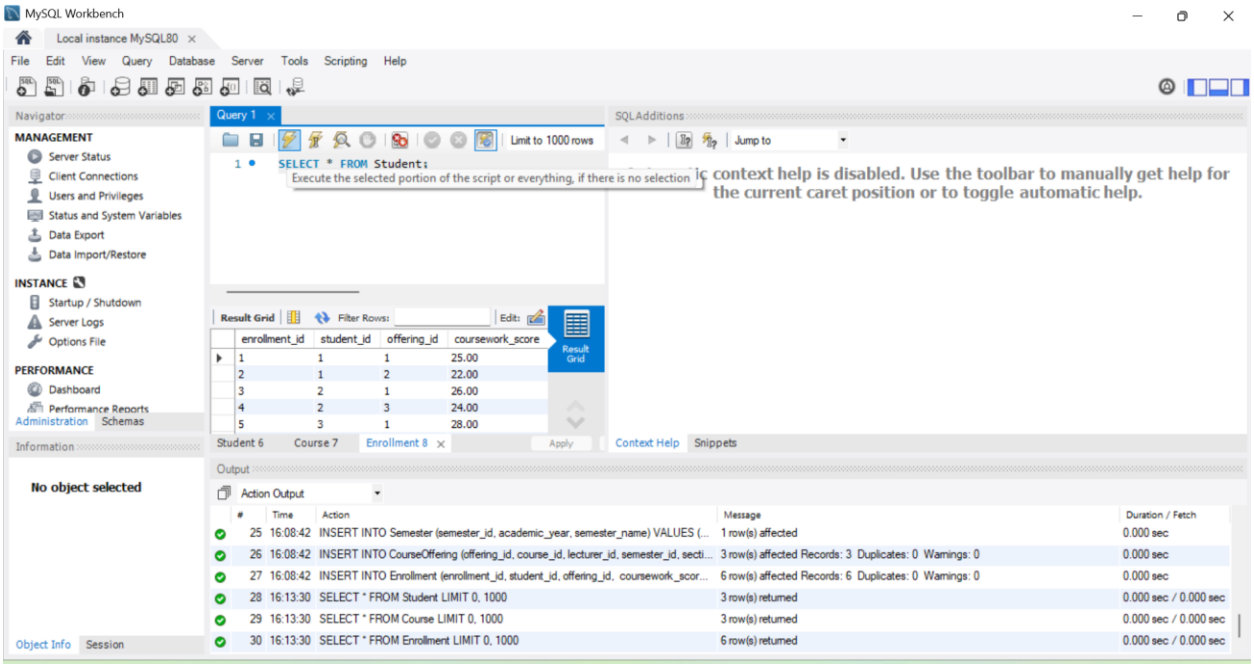
(4, 2, 3, 24, 62, 'Completed'),

(5, 3, 1, 28, 65, 'Completed'),

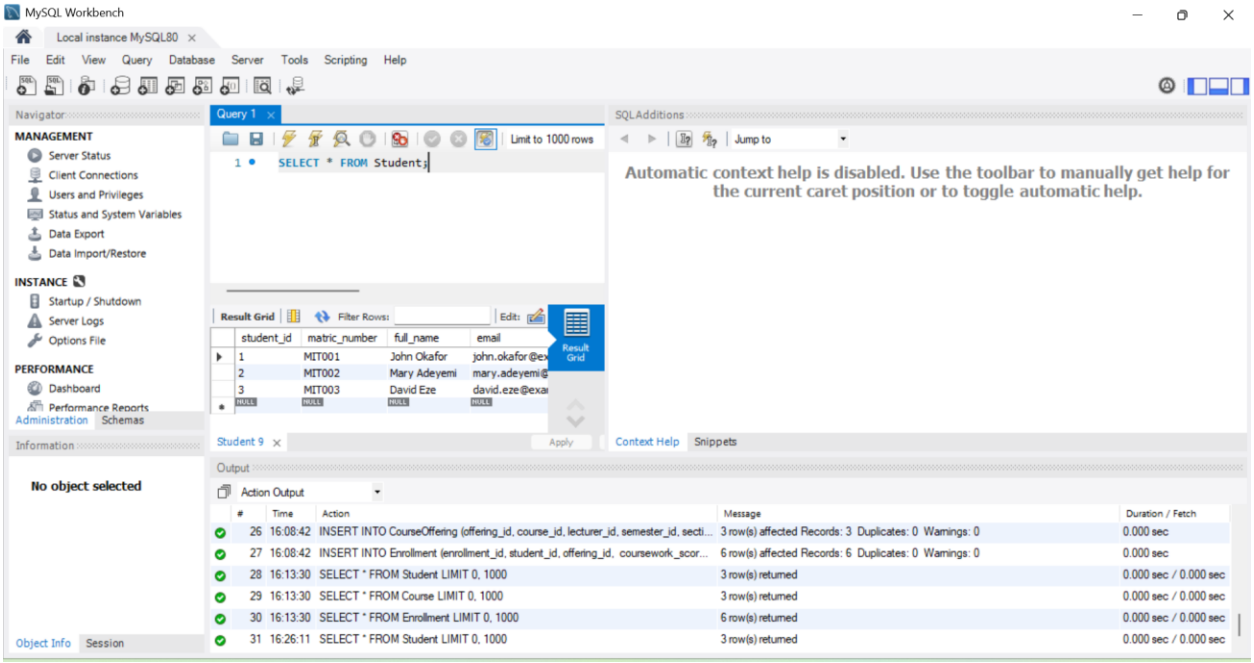
(6, 3, 2, 20, 50, 'Completed');



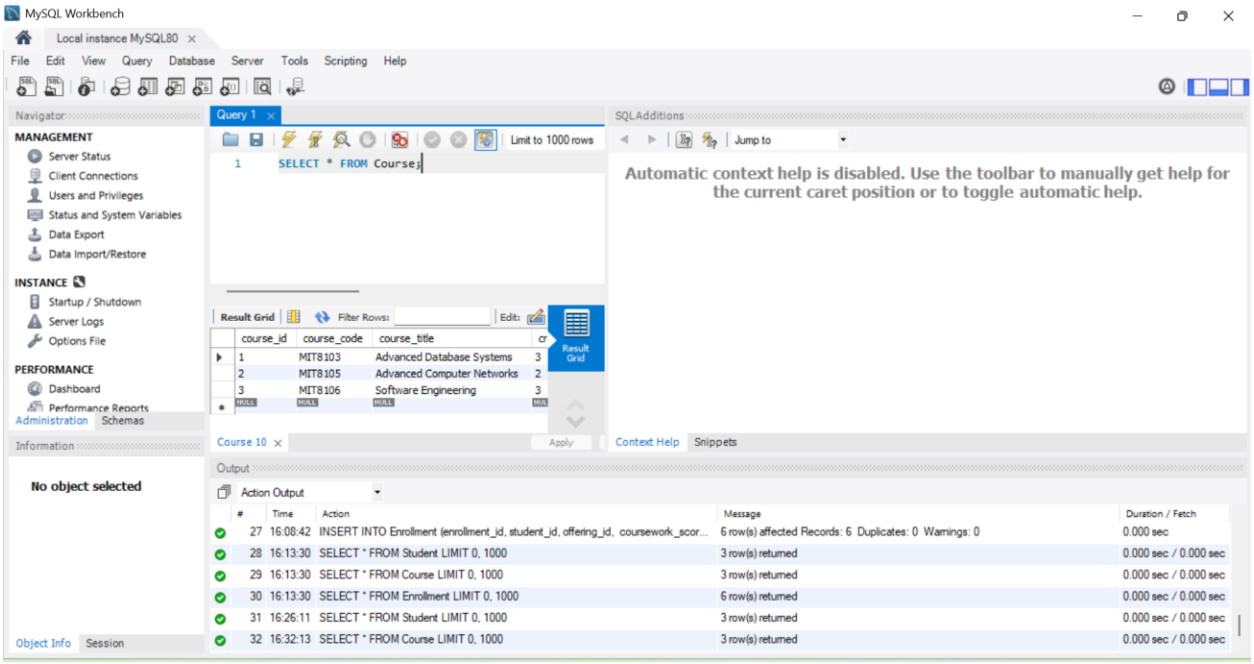

MySQL Workbench
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Performance Reports
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 SELECT * FROM Student;
2
3 SELECT * FROM Course;
4
5 SELECT * FROM Enrollment;
Result Grid
Filter Rows:
Edit:
Export/Import
Wrap Cell Content
Fetch rows:
enrollment_id student_id offering_id coursework_score
1 1 1 25.00
2 1 2 22.00
3 2 1 26.00
4 2 3 24.00
5 3 1 28.00
Student 6 Course 7 Enrollment 8
Apply
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
25 16:08:42 INSERT INTO Semester (semester_id, academic_year, semester_name) VALUES (... 1 row(s) affected 0.000 sec
26 16:08:42 INSERT INTO CourseOffering (offering_id, course_id, lecturer_id, semester_id, secti... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
27 16:08:42 INSERT INTO Enrollment (enrollment_id, student_id, offering_id, coursework_scor... 6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0 0.000 sec
28 16:13:30 SELECT * FROM Student LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
29 16:13:30 SELECT * FROM Course LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
30 16:13:30 SELECT * FROM Enrollment LIMIT 0, 1000 6 row(s) returned 0.000 sec / 0.000 sec


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
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Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Reports
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 SELECT * FROM Student;
Execute the selected portion of the script or everything, if there is no selection
Result Grid
Filter Rows:
Edit:
Export/Import
Wrap Cell Content
Fetch rows:
enrollment_id student_id offering_id coursework_score
1 1 1 25.00
2 1 2 22.00
3 2 1 26.00
4 2 3 24.00
5 3 1 28.00
Student 6 Course 7 Enrollment 8
Apply
SQLAdditions
Jump to
ic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
25 16:08:42 INSERT INTO Semester (semester_id, academic_year, semester_name) VALUES (... 1 row(s) affected 0.000 sec
26 16:08:42 INSERT INTO CourseOffering (offering_id, course_id, lecturer_id, semester_id, secti... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
27 16:08:42 INSERT INTO Enrollment (enrollment_id, student_id, offering_id, coursework_scor... 6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0 0.000 sec
28 16:13:30 SELECT * FROM Student LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
29 16:13:30 SELECT * FROM Course LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
30 16:13:30 SELECT * FROM Enrollment LIMIT 0, 1000 6 row(s) returned 0.000 sec / 0.000 sec

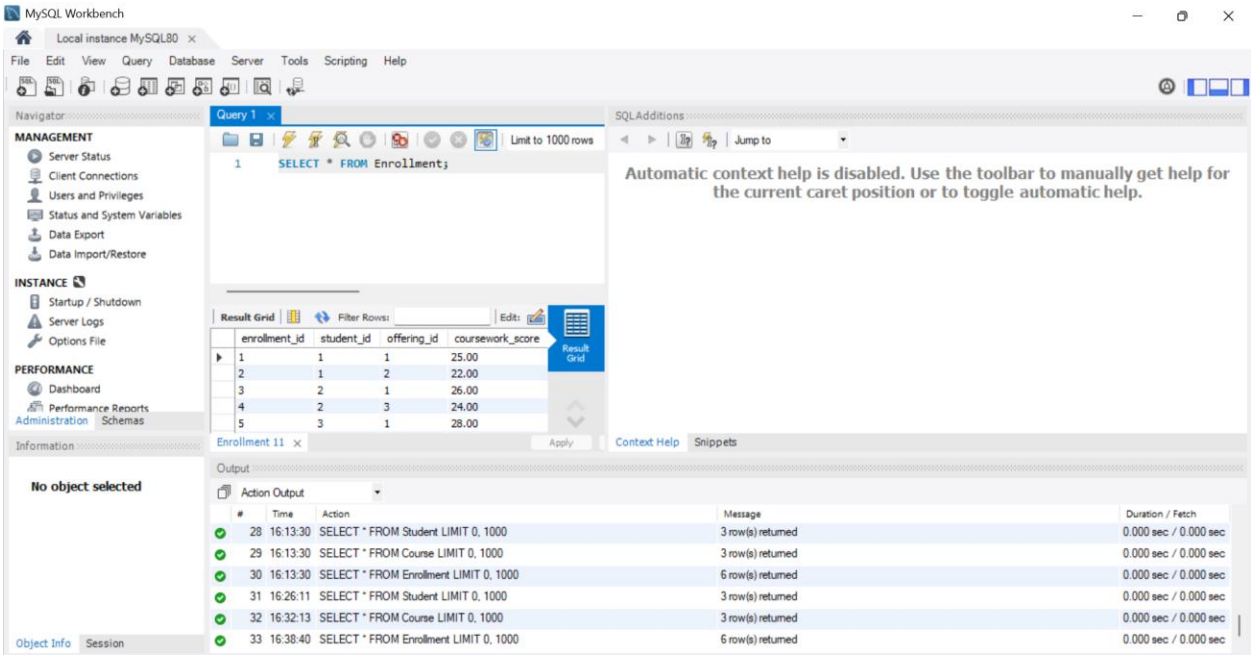
Verification of Student Table



Verification of Course Table



Verification of Enrollment Records



ER Diagram showing the Database Entities and their Relationships

